# Introduction to Machine Learning
# Worked Solutions and Marking Guide

Lectures 1–6 | Total: 100 marks

**Course:** ShanghaiTech CS182, 2026 Fall. This is an independent practice paper, not an official examination.

**Scope:** Introduction; Linear Algebra; Probability and Statistics; Optimization Primer; Bayesian Decision Theory; Parameter Estimation.

| Section | Questions | Marks each | Total |
|---|---|---|---|
| A. True/False | 20 | 1 | 20 |
| B. Multiple response | 20 | 2 | 40 |
| C. Proofs and worked problems | 8 | 5 | 40 |
| Total | 48 | | 100 |

**Scoring.** Section A: 1 mark for the correct T/F response; 0 otherwise. Section B: each item has four options and exactly one, two, or three correct options. Select all options you believe are correct. A complete correct set earns 2 marks; a nonempty proper subset of the correct set earns 1 mark; a response containing any incorrect option, or an empty response, earns 0. There is no negative marking. Section C: show reasoning; subpart marks sum to 5 for each problem. Valid alternative proofs and methods receive equivalent credit. **Quick answer key — Section A**

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| F | T | F | T | T | T | F | F | T | F |

| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|
| T | F | F | T | F | T | F | T | F | F |

**Quick answer key — Section B**

| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
|---|---|---|---|---|---|---|---|---|---|
| AC | C | ABD | B | ACD | BC | BD | CD | ACD | BCD |

| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
|---|---|---|---|---|---|---|---|---|---|
| A | ACD | BCD | AD | AB | BD | AB | ABC | D | C |

## Section A. Answers and Explanations

**1. False.** It is linear in the unknown coefficients $w$, although the basis functions are nonlinear in the input $x$. Linearity of a linear model refers to its parameters.
*Source: L1 p.21; L6 pp.42,90.*

**2. True.** The gradient stacks the $d$ partial derivatives. Its transpose times the $d$-component increment gives the scalar sum $\sum_j (\partial f/\partial w_j) dw_j$.
*Source: L2 pp.16–17; differential notation is a supplementary application.*

**3. False.** VC dimension uses the existence of a set that can be shattered. It does not require every configuration of four points to be shattered.
*Source: L1 p.41.*

**4. True.** The likelihood maximizer is $\widehat{\sigma}^2 = N^{-1} \sum_i (x_i - \bar{x})^2$. Divisor $N - 1$ instead corrects finite-sample bias; unbiasedness and likelihood maximization are different criteria.
*Source: L6 pp.16,19.*

**5. True.** For each fixed primal point, the Lagrangian is affine in the multipliers. A pointwise infimum of affine functions is concave, independently of primal convexity.
*Source: L4 p.21.*

**6. True.** The covariance terms vanish by independence, so the variance of the sum is $N\sigma^2$; dividing by $N^2$ gives the result.
*Source: L3 p.20; L6 p.20.*

**7. False.** For example, the uniform density on $(0, 1/4)$ equals four. The integral of a density is one; its pointwise values need not be bounded by one.
*Source: L3 pp.12,28.*

**8. False.** The posterior mean is optimal for squared-error loss when the second moment is finite. Absolute-error loss instead favors a posterior median; general losses may have other minimizers.
*Source: L6 p.25; absolute-loss comparison is supplementary.*

**9. True.** The best class has risk $1 - \max_i P(C_i \mid x)$, while rejection has risk $c$. Lower rejection cost therefore makes acceptance require greater posterior confidence.
*Source: L5 pp.17–18.*

**10. False.** Their intersection has probability zero, whereas independence would require the strictly positive product of their probabilities.
*Source: L3 pp.4,7.*

**11. True.** For every nonzero $v$, column independence implies $Av \neq 0$; hence $v^{\mathsf{T}} A^{\mathsf{T}} A v = \|Av\|_2^2 > 0$.
*Source: L2 pp.8,12–13.*

**12. False.** The invertible symmetric matrix $\mathrm{diag}(1, -1)$ is indefinite. Invertibility excludes zero eigenvalues but does not exclude negative ones.
*Source: L2 pp.9,12,14.*

**13. False.** Take $X$ uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Symmetry gives zero covariance, but $Y$ is determined by $X$ and is nonconstant. Joint Gaussianity is a special condition under which the implication does hold.
*Source: L3 pp.20,29; L6 p.58.*

**14. True.** For example, let a nonconstant binary class $C$ determine $X_1 = X_2 = C$. Within a class both features are constant and their conditional joint probabilities factor, but marginally the features coincide and are dependent.

*Source: L5 p.13; L6 pp.78,85–86; counterexample is supplementary.*

**15. False.** The exact result requires a normal population. For a general finite-variance population, a central-limit approximation is not an exact finite-sample t law. The normal-sample assumption is essential.

*Source: L3 pp.30,32–37; exact-normal assumption stated explicitly.*

**16. True.** Since $Q^\mathsf{T}Q = I$, $\|Qv\|_2^2 = v^\mathsf{T}Q^\mathsf{T}Qv = v^\mathsf{T}v$. Taking nonnegative square roots proves the statement.

*Source: L2 pp.6,10–11.*

**17. False.** Training error concerns a finite observed sample. Distinct hypotheses can agree on that sample and disagree elsewhere; zero training error alone gives no such guarantee.

*Source: L1 pp.39,48–50.*

**18. True.** Along any direction $v$, the restriction $t \mapsto f(x + tv)$ is convex and has second derivative $v^\mathsf{T}\nabla^2 f(x)v \geq 0$.

*Source: L4 pp.17–18.*

**19. False.** Inequality multipliers are nonnegative in this convention. Equality multipliers are unrestricted because multiplying a feasible equality by any real number still gives zero.

*Source: L4 pp.13–15.*

**20. False.** Complementary slackness gives $\lambda_i > 0 \Rightarrow f_i(x) = 0$, not the converse. For minimizing $x^2$ subject to $x \geq 0$, $x = 0$ is active and its KKT multiplier is zero.

*Source: L4 pp.11–12.*

## Section B. Option-by-Option Explanations

**21. Covariance parameter counts. Correct set: A, C.**

(A) **Correct.** Each symmetric three-by-three covariance has $3(4)/2 = 6$ free entries.

(B) **Incorrect.** All three variances are constrained to the same scalar, so only one parameter is free.

(C) **Correct.** Sharing removes the class multiplier, leaving six free entries.

(D) **Incorrect.** There are three variances per class, hence six in total.

*Source: L6 pp.74–75,78,80,82.*

**22. A quadratic objective. Correct set: C.**

(A) **Incorrect.** Its determinant is $16 - 4 = 12$, so its rank is two.

(B) **Incorrect.** The mixed second derivative of $2x_1x_2$ equals two, not one.

(C) **Correct.** The Hessian $\begin{bmatrix} 2 & 2 \\ 2 & 8 \end{bmatrix}$ is positive definite, and the gradient vanishes at the origin.

(D) **Incorrect.** Both leading principal minors of its Hessian are positive, so it is strictly convex.

*Source: L2 pp.12,18; L4 pp.18,22.*

**23. Derivatives of an affine function. Correct set: A, B, D.**

(A) **Correct.** Affine functions satisfy the convexity inequality with equality.

(B) **Correct.** Writing $f = (A^{\mathsf{T}}v)^{\mathsf{T}}w$ exposes the coefficient vector.

(C) **Incorrect.** The product $Av$ is not even defined: its inner dimensions would be three and two.

(D) **Correct.** The first derivative is constant in $w$.

*Source: L2 pp.17–18; L4 p.17.*

**24. Mahalanobis geometry. Correct set: B.**

(A) **Incorrect.** It is $2^2 + 1^2 = 5$; Euclidean distance does not divide by coordinate variances.

(B) **Correct.** The contributions are $2^2/4 = 1$ and $1^2/1 = 1$, so the squared distance is two.

(C) **Incorrect.** The distance is $\sqrt{2}$; confusing it with its square loses a square root.

(D) **Incorrect.** The semiaxes are two along the first coordinate and one along the second.

*Source: L6 pp.64–68.*

**25. A rectangular design matrix. Correct set: A, C, D.**

(A) **Correct.** The first two columns form the two-dimensional identity; two rows bound the rank above by two.

(B) **Incorrect.** Invertibility of the column Gram matrix requires full column rank, which a two-by-three matrix cannot have.

(C) **Correct.** It is a Gram matrix of rank two but size three by three.

(D) **Correct.** Multiplication gives $(0, 0)^{\mathsf{T}}$.

*Source: L2 pp.8,12–14.*

**26. Rejecting an uncertain prediction. Correct set: B, C.**

(A) **Incorrect.** The acceptance threshold is $1 - c = 0.70$, not $c$.

(B) **Correct.** Class risks are $0.54, 0.66, 0.80$, all greater than 0.30.

(C) **Correct.** Its posterior is largest and its conditional error is 0.54.

(D) **Incorrect.** It is the class-independent cost 0.30, after summing posterior probabilities.

*Source: L5 pp.16–18.*

**27. Two asymptotic statements. Correct set: B, D.**

(A) **Incorrect.** An asymptotic limit does not imply an exact finite-sample law.

(B) **Correct.** This is the weak law of large numbers.

(C) **Incorrect.** The CLT concerns normalized sums, not a change in the data-generating distribution of individual observations.

(D) **Correct.** This is the i.i.d. finite-variance central limit theorem.

*Source: L3 pp.23,30.*

**28. Polynomial models and validation. Correct set: C, D.**

(A) **Incorrect.** Lower training error can reflect fitting noise; it does not guarantee improved generalization.

(B) **Incorrect.** It can stay unchanged, for instance if the old model already fits the observations exactly.

(C) **Correct.** The degree is a complexity choice, and validation evaluates performance beyond the fitting sample.

(D) **Correct.** The larger class contains the smaller one; setting the extra coefficient to zero reproduces every old fit.

*Source: L1 pp.21–24,48–50; L6 pp.47,51.*

**29. Expectations and variances. Correct set: A, C, D.**

(A) **Correct.** Independence makes the covariance zero; the stipulated positive variances make correlation defined.

(B) **Incorrect.** In general the missing term is $2\,\mathrm{Cov}(X, Y)$.

(C) **Correct.** Linearity of expectation does not require independence.

(D) **Correct.** Translation does not affect variance and multiplication by three squares the scale.

*Source: L3 pp.18–20.*

**30. Step size in gradient descent. Correct set: B, C, D.**

(A) **Incorrect.** Convergence from a nonoptimal start requires $|1 - 4\eta| < 1$, namely $0 < \eta < 1/2$.

(B) **Correct.** The multiplier is $-7/5$, so its absolute value exceeds one.

(C) **Correct.** The multiplier is $-1$; the nonzero initial error keeps its magnitude.

(D) **Correct.** The error recurrence is $x_{t+1} - b = (1 - 4\eta)(x_t - b)$, whose factor is zero.

*Source: L4 p.24; convergence calculations are supplementary.*

**31. Base rates and evidence. Correct set: A.**

(A) **Correct.** The evidence is $0.6(0.2) + 0.1(0.8) = 0.20$, so the posterior is $0.12/0.20 = 0.6$.

(B) **Incorrect.** It is $1 - \max(0.6, 0.4) = 0.4$; 0.2 was the prior probability of $C_1$.

(C) **Incorrect.** Six is the likelihood ratio. Multiplying by prior odds 1/4 gives posterior odds 3/2.

(D) **Incorrect.** The two likelihoods must be weighted by their class priors, not simply added.

*Source: L3 pp.6,8; L5 pp.10–12.*

**32. Likelihood, MAP, and predictive integration. Correct set: A, C, D.**

(A) **Correct.** The logarithm is strictly increasing on positive values.

(B) **Incorrect.** Likelihood is a function of parameters with data fixed; it is not in general normalized over parameters.

(C) **Correct.** Multiplying the likelihood by a positive constant does not alter its maximizers.

(D) **Correct.** The posterior predictive averages the conditional prediction over parameter uncertainty.

*Source: L6 pp.9–10,22–24.*

**33. Shrinkage and estimation error. Correct set: B, C, D.**

(A) **Incorrect.** Its bias vanishes only at $\mu = 0$, so it is not an unbiased estimator over the whole parameter family.

(B) **Correct.** Multiplying the sample mean by one half scales its variance by one quarter.

(C) **Correct.** Apply the estimator MSE identity: variance plus squared bias.

(D) **Correct.** $E[T] - \mu = \mu/2 - \mu$.

*Source: L6 pp.17–20; shrinkage example is supplementary.*

**34. Dual functions. Correct set: A, D.**

(A) **Correct.** The Lagrangian can be unbounded below when minimized over unrestricted primal points.

(B) **Incorrect.** For minimization, the dual provides lower bounds: $d^* \leq p^*$.

(C) **Incorrect.** For example, minimizing $e^x$ over $\mathbb{R}$ has infimum zero and no minimizer; its unconstrained dual value is also zero.

(D) **Correct.** It is a pointwise infimum of affine functions of $\lambda$.

*Source: L4 pp.5–8,21.*

**35. A joint probability table. Correct set: A, B.**

(A) **Correct.** $E[XY] = 0.4$, $E[X] = 0.7$, and $E[Y] = 0.6$, giving $0.4 - 0.42 = -0.02$.

(B) **Correct.** Sum over $X$: $0.2 + 0.4 = 0.6$.

(C) **Incorrect.** It is $0.4/0.7 = 4/7$; the conditioning event is $X = 1$.

(D) **Incorrect.** For example, $p(1, 1) = 0.4 \neq 0.7 \times 0.6 = 0.42$.

*Source: L3 pp.13,17,20.*

**36. An exact normal-sample result. Correct set: B, D.**

(A) **Incorrect.** The degrees of freedom are $N - 1$, not $N$.

(B) **Correct.** This is an exact normal-sample property, used to obtain the Student t statistic.

(C) **Incorrect.** The residuals sum to zero and therefore are not $N$ independent standardized normal coordinates.

(D) **Correct.** Estimating the sample mean removes one degree of freedom from the centered sum of squares.

*Source: L3 pp.31–37.*

**37. Binary naive Bayes. Correct set: A, B.**

(A) **Correct.** Its class-weighted likelihood is $0.5 \times 0.8 \times (1 - 0.3) = 0.28$.

(B) **Correct.** A Bernoulli failure has probability $1 - p_{i2}$; zero is an observed value, not a missing feature.

(C) **Incorrect.** 0.56 is the class-conditional likelihood. Normalize weighted likelihoods 0.28 and 0.06, giving 14/17.

(D) **Incorrect.** Conditional independence requires a product: $0.4 \times (1 - 0.7) = 0.12$.

*Source: L5 pp.11–13; L6 p.85.*

**38. Bernoulli boundaries. Correct set: A, B, C.**

(A) **Correct.** Independence gives the Bernoulli variance divided by sample size.

(B) **Correct.** The likelihood is then $\theta^N$, increasing to its maximum at the boundary.

(C) **Correct.** The sample mean is unbiased, including the degenerate endpoint distributions.

(D) **Incorrect.** The specified closed parameter domain includes the boundary maximizer; stationarity is not necessary at a boundary.

*Source: L6 pp.11–12,19–20.*

**39. KKT with an active inequality. Correct set: D.**

(A) **Incorrect.** Here the optimum has objective derivative two, balanced by the constraint multiplier.

(B) **Incorrect.** Their stationarity residual is 2, not zero.

(C) **Incorrect.** The point $x = 0$ violates $x \geq 1$.

(D) **Correct.** They satisfy feasibility, $2x - \lambda = 0$, nonnegativity, and complementary slackness.

*Source: L4 pp.11–12,18,22.*

**40. Asymmetric loss. Correct set: C.**

(A) **Incorrect.** These are distinct loss criteria; the minimum risk here is 0.55.

(B) **Incorrect.** The cost four applies when the true class is $C_2$, so it multiplies 0.45.

(C) **Correct.** The two risks are $R(\alpha_1 \mid x) = 4(0.45) = 1.8$ and $R(\alpha_2 \mid x) = 0.55$.

(D) **Incorrect.** Maximum posterior is the special rule for equal unit error costs, not these asymmetric costs.

*Source: L5 pp.15–16.*

## 41. Quadratic forms and differentials

**Worked solution.** Write $A = S + K$, where $S = (A + A^{\mathsf{T}})/2$ and $K = (A - A^{\mathsf{T}})/2$. Then $S^{\mathsf{T}} = S$ and $K^{\mathsf{T}} = -K$. Since $w^{\mathsf{T}}Kw$ is a real scalar,

$$w^{\mathsf{T}}Kw = (w^{\mathsf{T}}Kw)^{\mathsf{T}} = w^{\mathsf{T}}K^{\mathsf{T}}w = -w^{\mathsf{T}}Kw.$$

Thus it is zero, proving (a).

For (b), $A$ is constant, so the product rule gives

$$df = \tfrac{1}{2}[(dw)^{\mathsf{T}}Aw + w^{\mathsf{T}}A\,dw] = \tfrac{1}{2}[(Aw)^{\mathsf{T}}dw + (A^{\mathsf{T}}w)^{\mathsf{T}}dw] = \left[\tfrac{1}{2}(A + A^{\mathsf{T}})w\right]^{\mathsf{T}} dw.$$

The first term is transposed using its scalar nature; the second uses $(A^{\mathsf{T}}w)^{\mathsf{T}} = w^{\mathsf{T}}A$. Comparing coefficients of the arbitrary increment $dw$ gives the claimed column gradient, of size $d \times 1$. The formula equals $Aw$ for every $w$ exactly when $A = A^{\mathsf{T}}$. For a particular $w$, equality can also occur if $(A^{\mathsf{T}} - A)w = 0$.

**Marking guide (5 marks).**

(a) 1 mark: symmetric and skew-symmetric decomposition; 1 mark: scalar-transpose argument and cancellation.

(b) 1 mark: differential product rule; 1 mark: correct transpose arrangement and gradient; 1 mark: symmetry condition, distinguishing equality for all vectors from accidental equality at one vector.

*Source: L2 pp.6,11,17; differential proof is supplementary.*

## 42. Maximum of affine functions

**Worked solution.** For arbitrary $x, y \in \mathbb{R}^d$ and $t \in [0,1]$, every index $i$ satisfies

$$a_i^{\mathsf{T}}(tx + (1-t)y) + b_i = t(a_i^{\mathsf{T}}x + b_i) + (1-t)(a_i^{\mathsf{T}}y + b_i) \le tF(x) + (1-t)F(y).$$

Taking the maximum over $i$ on the left proves the convexity inequality.

In (b), $F(x) = \max(x, -x) = |x|$. Its left derivative at zero is $-1$, and its right derivative is 1, so no derivative exists there. For distinct $x = 1$, $y = 3$, and $t = 1/2$,

$$F(2) = 2 = \tfrac{1}{2}F(1) + \tfrac{1}{2}F(3),$$

which violates the strict inequality required by strict convexity. A convex function need not have an ordinary Hessian everywhere. The Hessian test requires the differentiability conditions of the theorem; its absence does not establish nonconvexity.

**Marking guide (5 marks).**

(a) 1 mark: expand the affine functions at a convex combination; 1 mark: bound each term by the two maxima; 1 mark: take the maximum and conclude for all x,y,t.

(b) 1 mark: unequal one-sided derivatives; 0.5 mark: explicit equality for distinct points; 0.5 mark: correct limitation of the Hessian criterion.

*Source: L4 pp.17–18; finite-maximum construction is supplementary.*

## 43. A correlation bound and its equality case

**Worked solution.** Put $a = E[U^2] > 0$, $b = E[V^2] > 0$, and $k = E[UV]$. Expanding the nonnegative square gives

$$0 \le E[(U - tV)^2] = a - 2tk + t^2 b.$$

At $t = k/b$, this becomes $a - k^2/b \ge 0$, so $k^2 \le ab$. Dividing by $ab > 0$ yields $|\rho| \le 1$.

If $|\rho| = 1$, then $k^2 = ab$ and $E[(U - (k/b)V)^2] = 0$. A nonnegative random variable with zero expectation is zero almost surely, so $U = (k/b)V$ almost surely. Since $a > 0$, $k \neq 0$, and the constant is nonzero.

Conversely, if $U = cV$ almost surely with $c \neq 0$, then $a = c^2 b$ and $k = cb$. Therefore

$$\rho = \frac{cb}{\sqrt{c^2 b^2}} = \frac{c}{|c|}.$$

Thus $\rho = 1$ for $c > 0$ and $\rho = -1$ for $c < 0$. Almost-sure equality, rather than equality at just a few observed samples, is needed.

**Marking guide (5 marks).**

(a) 1 mark: expand the expected square; 1 mark: substitute t=k/b; 1 mark: normalize and conclude.

(b) 1 mark: zero expected square implies the almost-sure relation and nonzero coefficient; 1 mark: converse with sign computation.

*Source: L3 pp.19–20; L2 p.11; equality-case proof is supplementary.*

## 44. A certificate from weak duality

**Worked solution.** For any feasible point $u$, $\lambda_i f_i(u) \le 0$ and $\nu_j h_j(u) = 0$. Thus

$$g(\lambda, \nu) \le L(u, \lambda, \nu) \le f_0(u).$$

Since this holds for every feasible $u$, taking their infimum gives $g(\lambda, \nu) \le p^*$. By the definition of an infimum, $p^* \le f_0(x)$ at every feasible $x$. Neither attainment nor a zero optimal duality gap was used.

For (b), the established bounds give

$$0 \le f_0(x) - p^* \le f_0(x) - g(\lambda, \nu) \le \epsilon.$$

This is a certificate on the objective-value error, not automatically on the distance between $x$ and an optimizer. Equality multipliers are unrestricted because their contributions vanish at every feasible point regardless of sign.

**Marking guide (5 marks).**

(a) 1 mark: compare g with L at a feasible point; 1 mark: establish $L \le f_0$ using the correct signs; 1 mark: take the infimum and give both bounds without assuming attainment.

(b) 1 mark: epsilon inequality chain; 1 mark: explain unrestricted equality multipliers and the objective-value interpretation.

*Source: L4 pp.6–10,13–14; epsilon certificate is supplementary.*

## 45. Three classes, unequal losses, and rejection

**Worked solution.** The prior-weighted likelihoods are $(0.12, 0.10, 0.12)$. Hence

$$p(x) = 0.34 = \frac{17}{50}, \qquad (P(C_1 \mid x), P(C_2 \mid x), P(C_3 \mid x)) = \left(\frac{6}{17}, \frac{5}{17}, \frac{6}{17}\right).$$

With rows as actions, the risks are

$$R_1 = \frac{2(5) + 4(6)}{17} = 2, \quad R_2 = \frac{3(6) + 1(6)}{17} = \frac{24}{17}, \quad R_3 = \frac{1(6) + 5(5)}{17} = \frac{31}{17}.$$

The rejection risk is $6/5$, lower than every class risk, so reject. If the rejection cost becomes $7/5$, it is still slightly below $24/17$: $7 \cdot 17 = 119 < 120 = 24 \cdot 5$. Therefore rejection remains optimal. The near comparison requires exact values, not rounding both risks to 1.4. The most probable classes are $C_1$ and $C_3$, whereas $C_2$ has the lowest class-action risk. Posterior ranking ignores the unequal losses. The $1 - c$ acceptance threshold derived for unit error costs does not apply to this loss matrix; compare the actual conditional risks.

**Marking guide (5 marks).**

(a) 1 mark: evidence from weighted likelihoods; 1 mark: all normalized posteriors.

(b) 1.5 marks: three class risks (0.5 each); 0.5 mark: rejection risk and action.

(c) 0.5 mark: correct exact comparison at 7/5; 0.5 mark: explain posterior/risk mismatch.

*Source: L5 pp.11–12,15–18.*

## 46. Normal observations with a normal prior

**Worked solution.** Up to factors independent of $\theta$,

$$p(\theta \mid \mathcal{D}) \propto \exp\left\{-\frac{1}{2}\left[\frac{\sum_i (x_i-\theta)^2}{\sigma^2}+\frac{(\theta-\mu_0)^2}{\tau_0^2}\right]\right\}.$$

Expanding the bracket produces a quadratic coefficient $a = N/\sigma^2 + 1/\tau_0^2$ and a linear term $-2b\theta$, where $b = \sum_i x_i/\sigma^2 + \mu_0/\tau_0^2$. Completing the square gives

$$a\theta^2 - 2b\theta + \text{constant} = a(\theta - b/a)^2 + \text{constant}.$$

Therefore the posterior is $\mathcal{N}(m_N, v_N)$, with

$$v_N = \left(\frac{N}{\sigma^2}+\frac{1}{\tau_0^2}\right)^{-1}, \qquad m_N = v_N\left(\frac{N\bar{x}}{\sigma^2}+\frac{\mu_0}{\tau_0^2}\right).$$

Numerically, $\bar{x} = 4$, posterior precision is $3/4 + 1/2 = 5/4$, and

$$v_N = \frac{4}{5}, \qquad m_N = \frac{4}{5}\left(3+\frac{1}{2}\right) = \frac{14}{5}.$$

The normal posterior has coincident mean and mode, so $\theta_{\text{MAP}} = 14/5$. For prediction, conditional expectation gives $E[X_{\text{new}} \mid \mathcal{D}] = m_N = 14/5$. Write $X_{\text{new}} = \theta + \varepsilon$, where the new noise has variance four and is independent of $\theta$ and the data. Then

$$\text{Var}(X_{\text{new}} \mid \mathcal{D}) = 4 + v_N = \frac{24}{5}.$$

The extra $4/5$ is residual uncertainty about $\theta$; a plug-in estimate would omit it.

**Marking guide (5 marks).**

(a) 1 mark: posterior kernel; 1 mark: quadratic/linear coefficients and completing square; 1 mark: both general posterior parameters.

(b) 0.5 mark: numerical mean and variance; 0.5 mark: MAP equals the posterior mean.

(c) 0.5 mark: predictive mean and variance; 0.5 mark: identify parameter uncertainty as the additional variance.

*Source: L6 pp.22–27; predictive variance is a supplementary application.*

## 47. A Gaussian decision boundary

**Worked solution.** The Gaussian normalization and the common quadratic term cancel between classes. In general,

$$g_2 - g_1 = (\mu_2 - \mu_1)^\top \Sigma^{-1} x - \frac{1}{2}(\mu_2^\top \Sigma^{-1} \mu_2 - \mu_1^\top \Sigma^{-1} \mu_1) + \log \frac{P(C_2)}{P(C_1)}.$$

Here $\Sigma^{-1} = \mathrm{diag}(1/2, 1)$, $\Sigma^{-1}\mu_2 = (1, 1)^\top$, and $\mu_2^\top \Sigma^{-1} \mu_2 = 3$. Thus

$$g_2(x) - g_1(x) = x_1 + x_2 - \frac{3}{2} + \log 3.$$

The boundary is $x_1 + x_2 = 3/2 - \log 3$. Predict $C_2$ on and above this line, and $C_1$ below it, using the stated tie convention.

At $x_A$, the difference is $-3/2 + \log 3 < 0$, so predict $C_1$. At $x_B$, it is $-1/2 + \log 3 > 0$, so predict $C_2$. The unequal priors shift the boundary relative to the equal-prior threshold $3/2$.

A full symmetric two-by-two covariance has three free parameters. Two separate covariances require six; one shared full covariance requires three. Means and priors are excluded from this count. The diagonal matrix specified in the original numerical model would require only two covariance parameters if its diagonal structure were imposed during estimation.

**Marking guide (5 marks).**

(a) 1 mark: cancel common Gaussian terms; 1 mark: correct coefficients and prior term; 1 mark: boundary and correctly oriented regions with tie rule.

(b) 0.5 mark for each correctly classified point with sign justification.

(c) 0.5 mark: six separate covariance parameters; 0.5 mark: three shared full-covariance parameters.

*Source: L6 pp.70,75–76,82.*

## 48. A constrained quadratic and its dual

**Worked solution.** Use the equality to set $v = -u$. The reduced objective is

$$\frac{1}{2}[(u-2)^2 + (1-u)^2] = u^2 - 3u + \frac{5}{2}, \quad u \le 1.$$

Its unconstrained minimizer is $u = 3/2$, outside the feasible half-line. Its derivative $2u - 3$ is negative for $u \le 1$, so the constrained minimum occurs at $u^* = 1$, $v^* = -1$, with objective $p^* = 1/2$. Strict convexity on a convex feasible set makes this minimizer unique.

Stationarity requires

$$u - 2 + \lambda + \nu = 0, \qquad v + 1 + \nu = 0.$$

At the optimizer this gives $\nu^* = 0$ and $\lambda^* = 1$. Primal feasibility holds since $1 - 1 = 0$ and $1 + (-1) = 0$. Dual feasibility holds since $1 \ge 0$; complementary slackness is $1(1-1) = 0$; the two stationarity residuals are zero. The objective Hessian is $I$, and the constraints are affine, so these KKT conditions also suffice for global optimality.

For the dual calculation, expand the Lagrangian as

$$L = \tfrac{1}{2}u^2 + (-2 + \lambda + \nu)u + \tfrac{1}{2}v^2 + (1 + \nu)v + \tfrac{5}{2} - \lambda.$$

Minimizing each square gives

$$g(\lambda, \nu) = \frac{5}{2} - \lambda - \frac{1}{2}(-2 + \lambda + \nu)^2 - \frac{1}{2}(1 + \nu)^2.$$

Therefore $g(1, 0) = 1/2 = p^*$, and the primal–dual objective gap is zero.

**Marking guide (5 marks).**

(a) 1 mark: reduced objective and feasible minimizer; 1 mark: value and strict-convexity uniqueness argument.

(b) 1 mark: correct multipliers from stationarity; 1 mark: primal/dual feasibility, complementary slackness, and both stationarity checks.

(c) 0.5 mark: correct general dual function; 0.5 mark: value 1/2 and zero gap.

*Source: L4 pp.4–8,11–15,20.*

## Source Scope and Authoring Notes

The primary sources are the six official PDFs in `E:\CS182\2026fall\lecture`. Page numbers count the cover as PDF page 1. Course Overview is not counted as one of the six lectures. L1–L6 denote the following files.

**L1** Introduction and supervised learning: `Introduction.pdf` (52 pages).

**L2** Linear algebra: `Linear_Algebra.pdf` (18 pages).

**L3** Probability and statistics: `Probability_Statistics.pdf` (39 pages).

**L4** Optimization primer: `Optimization_Primer.pdf` (25 pages).

**L5** Bayesian decision theory: `Bayesian_Decision_Theory.pdf` (23 pages).

**L6** Parameter estimation: `Parameter_Estimation.pdf` (90 pages).

Questions are newly composed for this paper. The written homework files in the repository were consulted only for the style of explicit assumptions, linked subparts, derivations, and point allocations. No original homework problem, original dataset, or original requested proof was reused as a long question. In particular, the long questions do not ask for the homework proofs of covariance positive semidefiniteness, the norm convexity theorem, the original ridge normal equations, or the exponential–Gamma posterior.

Supplementary applications of lecture tools are labeled in individual source notes. They use concepts and methods within this scope and supply any extra assumptions or definitions needed. Exact Student t results explicitly assume a normal population; marginal uncorrelatedness is not silently treated as independence. A posterior mean is claimed optimal only under squared-error loss, and a constrained optimum need not have zero objective gradient.

Coverage is intentionally weighted toward common conceptual errors in Sections A and B. Section C contains four mathematical proof problems (41–44), followed by Bayesian decisions, posterior derivation and prediction, Gaussian classification, and constrained optimization (45–48).